STAT3888: Causal Inference and DAG Analysis

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1 Terminology and Purpose

The purpose of this note is to use causal inference concepts to justify which variables should be adjusted for in our analysis of physical activity and liver biomarkers. Rather than including covariates solely because they are available, a causal **directed acyclic graph (DAG)** allows us to state our assumptions about how the variables relate and identify an appropriate adjustment set.

Table 1: Causal Inference Terminology and Guidelines

Term	Intuitive meaning	What we usually do
DAG	Diagram representing assumed causal relationships between variables using directed arrows	Use it to reason about which variables should be adjusted for
Confounder	A variable that affects both the predictor and outcome, e.g., $X \leftarrow Z \rightarrow Y$	Usually adjust for it
Mediator	A variable lying on the causal pathway from predictor to outcome, e.g., $X \rightarrow M \rightarrow Y$	Do not adjust if estimating the **total effect**
Collider	A variable caused by two other variables, e.g. $X \rightarrow C \leftarrow Y$	Usually do **not** condition on it
Proxy	An observed variable used as an imperfect stand-in for another variable that is difficult or impossible to measure directly	May be useful, but interpretation depends on what it is proxying
Latent variable	An underlying variable that is not directly observed	May sometimes be partially captured by proxies
Backdoor path	A non-causal path between predictor and outcome that enters the predictor through an incoming arrow, e.g., $X \leftarrow \ldots \rightarrow Y$	Block these paths by adjusting for appropriate confounders
Adjustment set	A collection of variables sufficient to block relevant backdoor paths	These become the covariates used for causal adjustment

2 Variable Roles

We will list out our main variables of interest, as well as my current proposed causal classification.

Table 2: Variable Roles and Causal Classifications

Variable	Role in analysis	Proposed causal classification
`EXREGUIC`	Main predictor/exposure	**Exposure**
ALT / GGT	Response	**Outcome**
`AGEB`	Adjustment variable	**Confounder**
`SEX`	Adjustment variable	**Confounder**
`WAIST`	Adjustment variable	**Confounder or mediator**
`DSMKHHRB`	Adjustment variable	**Potential confounder / proxy**
Smoking status	Population restriction	**Selection / subgroup variable**
Alcohol-risk category	Population restriction	**Selection / subgroup variable**

There are some important considerations here, which I order in terms of importance. First, WAIST can either be a confounder or a mediator, which is a more important distinction here as this classification directly determines our model covariates. Hence, we require some domain understand on how physical activity, waist circumference, and liver biomarkers interact.

Second, the number of daily smokers (DSMKHHRB) can be treated as both a confounder or a proxy. It may be worth determining whether it is a proxy for another variable such as socioeconomic status (SES), but in either case we should adjust for it in the model.

Finally, the smoking status and alcohol-risk category are selection variables, as our problem explicitly considers a particular sub-population regarding these variables. However, conditioning on these variables changes the population to which the estimated effect applies. Depending on the underlying causal structure, such selection may also introduce selection bias, but this requires additional assumptions and cannot be concluded solely from the fact that the sample was restricted.

3 Candidate Causal Structures and Adjustment Sets

I present the two DAGs I believe can portray our causal structure, depending on whether WAIST is treated as a mediator or a confounder.

3.1 DAG 1: WAIST as a Mediator

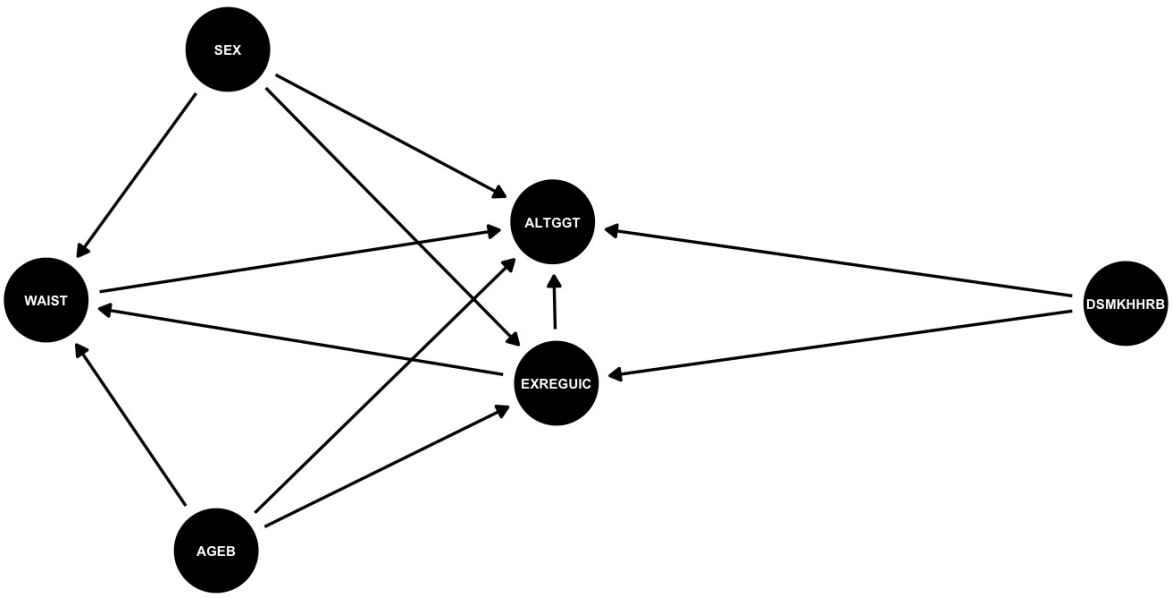


Figure 1: DAG 1: WAIST as a Mediator.

In this DAG, physical activity affects waist circumference, which subsequently affects ALT/GGT. In other words, we assume that

$$\texttt{EXREGUIC} \longrightarrow \texttt{WAIST} \longrightarrow \texttt{ALT/GGT}.$$

Therefore waist lies on a causal pathway from activity to liver biomarkers. For the *total effect* of physical activity, we **should not adjust for** WAIST, as otherwise we risk underestimating the effect of our predictor. Then, using `dagitty::adjustmentSets()` we find the adjustment set to be $\{\texttt{AGEB}, \texttt{SEX}, \texttt{DSMKHHRB}\}$.

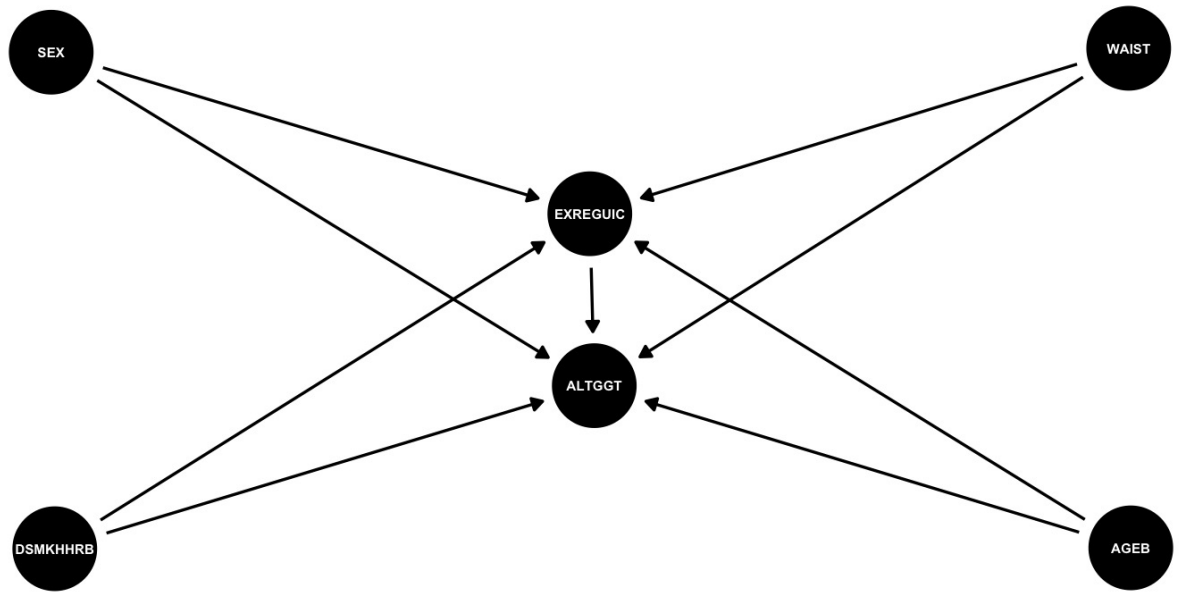


Figure 2: DAG 2: WAIST as a Confounder.

3.2 DAG 2: WAIST as a Confounder

In this DAG, waist circumference influences both the likelihood of meeting physical activity guidelines and ALT/GGT. In other words, we assume that

$$\texttt{EXREGUIC} \longleftarrow \texttt{WAIST} \longrightarrow \texttt{ALT/GGT}$$

which is a backdoor path, and therefore waist *should be adjusted for*. Using `dagitty::adjustmentSets()` we find the adjustment set to be $\{\texttt{AGEB}, \texttt{SEX}, \texttt{WAIST}, \texttt{DSMKHHRB}\}$. This is what we currently have in our models.

4 Statistical Implications

The principal benefit of the DAG analysis is not the creation of a new predictive model, but justification of which covariates should be included when estimating the association between physical activity and ALT/GGT.

Under the confounder interpretation of waist, the existing model is appropriate:

$$\texttt{ALT/GGT} \sim \texttt{EXREGUIC} + \texttt{AGEB} + \texttt{SEX} + \texttt{WAIST} + \texttt{DSMKHHRB}.$$

Under the mediator interpretation, adjustment for waist may remove part of the total effect of physical activity, suggesting the alternative specification:

$$\texttt{ALT/GGT} \sim \texttt{EXREGUIC} + \texttt{AGEB} + \texttt{SEX} + \texttt{DSMKHHRB}.$$

5 Unresolved Matters

The largest uncertainty in the current DAG is the temporal and causal relationship between physical activity and waist circumference. There are two plausible mechanisms:

1. Higher waist $\longrightarrow$ lower likelihood of exercise, making `WAIST` a *confounder*.
2. Exercise $\longrightarrow$ reduced waist $\longrightarrow$ ALT/GGT, making `WAIST` a *mediator*.

Because the available data are cross-sectional, temporal ordering cannot be established directly from the dataset. Domain knowledge and relevant literature are therefore required.

Questions requiring domain input

1. Is waist circumference more appropriately treated as a pre-existing determinant of physical activity, a consequence of long-term physical activity, or both?
2. Is there literature supporting a dominant causal direction between physical activity, waist circumference, and ALT/GGT?
3. Is household smoking exposure (`DSMKHHRB`) plausibly related to both physical activity and liver biomarkers?
4. Could `DSMKHHRB` reasonably act as a proxy for broader household/environmental or socioeconomic factors?

5.1 Proposed Next Steps

1. Validate the DAG assumptions with the nutrition team.
2. Determine whether waist should primarily be treated as a confounder or mediator.
3. Fit both model specifications if the waist role remains uncertain.
4. Compare the estimated `EXREGUIC` effect between:
   - adjustment with waist, and
   - adjustment without waist.
5. Treat this comparison as a causal sensitivity analysis.